



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE

(AUTONOMOUS)

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)

Accredited by NAAC with 'A+' Grade

Recognised as Scientific and Industrial Research Organisation

SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Regulation: R23									
MINOR IN QUANTUM TECHNOLOGIES									
(Applicable for CSE, AIML, CIC, IT, AIDS, CSBS, CSD, CSIT, CE, ECE, EEE & ME)									
COURSE STRUCTURE									
(With effect from 2023-24 admitted Batch onwards)									
Course Code	Course Name	Year/Sem	Cr	L	T	P	C.I.E	S.E.E	Total Marks
B23QTM101	Survey of Quantum Technologies and Applications	II-II	3	3	0	0	30	70	100
B23QTM102	Foundations of Quantum Technologies	III-I	3	3	0	0	30	70	100
B23QTM103	Basic Programming Lab	III-I	3	2	0	2	30	70	100
B23QTM104	Introduction to Quantum Computation	III-II	3	3	0	0	30	70	100
B23QTM105	Introduction to Quantum Communication	IV-I	3	3	0	0	30	70	100
B23QTM106	MOOCS course		3	3	0	0	30	70	100
TOTAL			18	17	0	2	180	420	600

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23QTM101	Minors	3	--	--	3	30	70	3 Hrs.
SURVEY OF QUANTUM TECHNOLOGIES AND APPLICATIONS								
(Minor Degree Course in Quantum Technologies)								
Course Objectives: The main objectives of the course is to								
1.	Provide an overview of the fundamental principles and concepts underlying quantum technologies and their four main verticals.							
2.	Familiarize students with the current state-of-the-art developments, platforms, and use-cases across all four quantum technology verticals.							
3.	Highlight exciting and emerging applications of quantum technologies in science, engineering, and industry, without focusing on rigorous mathematical depth.							
4.	Develop students' ability to relate classical intuition to quantum ideas, enabling them to interpret qualitative descriptions of quantum systems and protocols.							
5	Motivate students to explore advanced coursework or research in specific quantum technology verticals by exposing them to contemporary challenges and open problems.							
Course Outcomes: At the end of the course Students will be able to,								
S. No	Outcome							Knowledge Level
1.	Explain the general physical principles underlying the realization of qubits for computation.							K2
2.	Identify various hardware implementations of qubits used for quantum computation.							K3
3.	Describe the basic ideas and concepts of quantum sensing.							K2
4.	Illustrate the applications of quantum sensing in different domains.							K3
5.	Explain the implementation of quantum communication protocols over fibre-based and free-space channels.							K2
SYLLABUS								
UNIT-I (08 Hrs)	Quantum Technologies – four verticals: Motivation for Quantum Technologies A qualitative overview of salient aspects of quantum physics: Quantum States, Wave functions, Probabilistic interpretation, Physical observables, Hermitian operators, expectation values, Heisenberg uncertainty principle, Schrodinger equation, Time evolution; distinction from classical physics; Heuristic description of Superposition, Tunneling and entanglement; No cloning theorem; Simulating classical systems – Feynman's idea of a quantum simulator and the birth of the field.							
UNIT-II (10 Hrs)	Quantum Computation: Basics of qubits -- what is a qubit? How is it different from a classical bit? – Review of classical logic gates; Di Vincenzo criteria for realizing qubits; Basics of qubit gates and quantum circuits; Physical implementation of qubits (very qualitative description); Solid State Qubits: Semiconducting Qubits – quantum dots,							

	spins, Superconducting Qubits – charge, flux and phase, Topological Qubits – proposals and advantages; Atoms and Ions: Trapped ions, Rydberg atoms, Neutral atoms; Photonic Qubits: Conventional linear optical setups, Integrated Photonics; NMR qubits: Conventional NMR qubits, NV centres Overview of applications and recent achievements: RSA and Shor’s algorithm, Quantum Advantage; Long term goals and strategies being followed : Error correction
UNIT-III (12 Hrs)	Quantum Sensing: Basics of quantum sensing, Basics of Photon (single and entangled) generation and detection, Gravimetry, Atomic clock, Magnetometry, State of the art in Quantum Sensing
UNIT-IV (10 Hrs)	Quantum Communications: Basics of digital communication, Quantifying classical information – Shannon entropy, Basic ideas of quantum communication, security, eavesdropping, Overview of quantum communication achievements : Terrestrial – fibre-based, Free space, Satellite-based
UNIT-V (10 Hrs)	Introduction to Quantum Materials: What are quantum materials, Why are they important, Applications (quantum computing, spintronics, etc.) Overview of Key Classes of Quantum Materials: Topological Insulators, Superconductors, Mott Insulators, 2D Materials and Quantum Spin Liquids.
Textbooks:	
1.	Quantum Information Science – Manenti R., Motta M., 1st Edition, Oxford University Press (2023)
2.	Quantum computation and quantum information – Nielsen M. A., and Chuang I. L., 10th Anniversary edition, Cambridge University Press (2010)
Reference Books:	
1.	Elements of Quantum Computation and Quantum Communication, A. Pathak, Boca Raton, CRC Press (2015)
2.	An Introduction to Quantum Computing, Phillip Kaye, Raymond Laflamme, and Michele Mosca, Oxford University Press (2006)
3.	Quantum computing explained, David McMahon, Wiley (2008)
e-Resources	
1.	Learn quantum computing, IBM Quantum Learning
2.	Introduction to Quantum Computing, nptel.ac.in/courses/106106232

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23QTM102	Minors	3	--	--	3	30	70	3 Hrs.
FOUNDATIONS OF QUANTUM TECHNOLOGIES								
(Minor Degree Course in Quantum Technologies)								
Course Objectives: The main objectives of the course is to								
1.	Provide a rigorous understanding of the central theoretical concepts of quantum mechanics.							
2.	Equip students with various analytical and mathematical techniques used in quantum physics							
3.	Foster strong physical intuition for interpreting and solving quantum mechanical problems.							
Course Outcomes: At the end of the course students will be able to								
S.No	Outcome							Knowledge Level
1.	Apply the most relevant mathematical techniques to quantum physics problems and analyze their implications.							K3
2.	Explain and evaluate the basic postulates of quantum mechanics, as well as their applications in physical systems.							K2
3.	Analyze foundational concepts and solve problems in Statistical Physics.							K3
4.	Integrate basics of Information Science to describe and interpret quantum systems.							K3
5.	Assess the basics of computational complexity as it relates to quantum computing scenarios.							K2
SYLLABUS								
UNIT-I (10 Hrs)	Brief overview of classical physics (This segment is meant for the student to understand what a Hamiltonian is, which will feature later in quantum mechanics) :Hamiltonian function and Hamilton’s equations, Phase-space description of a system, Connection and Equivalence with Newton’s laws for simple systems – free particle, particle moving in a conservative potential, examples of Harmonic oscillator, hydrogen atom; Historical evolution of quantum mechanics: Planck’s quantum hypothesis, Photo electric effect, Atomic spectra, Bohr’s quantisation principle, De Broglie’s Wave particle duality.							
UNIT-II (10 Hrs)	Postulates of Quantum Mechanics: State vectors and Hilbert Space, Dirac Bra-Ket notation, Measurables and Hermitian Operators, Unitary Transformations, Schrodinger Equation and Time evolution of quantum states, Measurement Postulate, Schrodinger, Heisenberg and Interaction pictures, Eigen values, Expectation values and Matrix elements, Heisenberg’s Uncertainty principle							
UNIT-III (10 Hrs)	Density operator formalism of quantum mechanics – pure and mixed states; Superposition and Entanglement in quantum mechanics; No cloning theorem; Applications of postulates – Particle in a box, Hydrogen atom, Harmonic Oscillator Number states, ladder operators and Coherent states of a harmonic oscillator; Spin and Angular momentum – spin half particles; Rabi problem of a spin-half particle in a							

	rotating magnetic field; Bosons and Fermions.
UNIT-IV (10 Hrs)	Statistical Physics: Quick review of first and second laws of thermodynamics, Thermal Equilibrium and Gibbs principle, Applying Gibbs principle to Classical and Quantum harmonic oscillators, Bosons and Fermions and Quantum statistics – Fermi-Dirac and Bose- Einstein distributions
UNIT-V (10 Hrs)	Information Science: Digital communication and information: Quantifying information in terms of Shannon entropy; Basic ideas of quantum information; Decoherence and noise; Introductory ideas of Kraus operators Brief overview of Computational Complexity: Qualitative ideas of a Turing machine: Types of Turing machines; Time and Space complexity – P vs NP, PSPACE; Quantum complexity classes – Q, EQP, BQP, BPP, QMA; Post Quantum Cryptography (PQC)
Textbooks:	
1.	Introduction to Quantum Mechanics, Griffiths D. J., 3rd Edition, Cambridge University Press (2024)
2.	Introduction to Electrodynamics, Griffiths D. J., 4th edition, Cambridge University Press (2020)
Reference Books:	
1.	Principles of Quantum Mechanics, Shankar, R., 2nd edition, Springer (2014)
2.	Quantum Information Science – Manenti R., Motta M., 1st Edition, Oxford University Press (2023)
3.	Quantum computation and quantum information – Nielsen M. A., and Chuang I. L., 10 th Anniversary edition, Cambridge University Press (2010)
4.	A Pathak, Elements of Quantum Computation and Quantum Communication, Boca Raton, CRC Press (2015)
5.	Information Theory, Robert B. Ash, Dover Publications (2003)
6.	Statistical Mechanics, Pathria R. K., Paul D. Beale, 4th edition, Academic Press, (2021)
e-Resources	
1.	Quantum Physics I, Quantum Physics I Physics MIT OpenCourseWare
2.	Quantum Physics II, Quantum Physics II Physics MIT OpenCourseWare
3.	Statistical Physics I, Statistical Physics I Physics MIT OpenCourseWare
4.	Quantum Information and Computation, https://www.lorentz.leidenuniv.nl/quantumcomputers/literature/preskill_1_to_6.pdf

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23QTM103	Minor	2	0	2	3	30	70	3 Hrs.
BASIC PROGRAMMING LAB								
(Minor Degree Course in Quantum Technologies)								
Pre-requisites: Basic Physics, Linear Algebra, and Introduction to Modern Physics.								
Course Objectives: Students are expected								
1.	This course is meant to provide students a quick hands-on experience in scientific computing and its applications to areas within Quantum Technologies.							
Course Out Comes: At the end of the course students will be able to								
S. No	OUT COME							Knowledge Level
1.	Apply programming basics to solve scientific problems.							K3
2.	Implement techniques for scientific computing.							K3
3.	Demonstrate applications of scientific computing in quantum mechanics.							K4
4.	Analyze applications of scientific computing in electromagnetism.							K4
SYLLABUS								
UNIT-I (10Hrs)	Basics of programming ◦ Data structures, classes, Object-oriented programming ◦ Data storage and retrieval, Memory allocation ◦ Scientific plotting, documentation of codes Simple algorithms and benchmarking run time ◦ Sorting ◦ Searching ◦ Arithmetic algorithms like GCD, Prime factorisation							
UNIT-II (10 Hrs)	Numerical Integration and differential equations ◦ Linear 2nd Order ODEs with constant coefficients ◦ Linear 2nd order ODEs with variable coefficients ◦ Boundary value problems - Poisson equation - Laplace equation - Wave equation - Diffusion Equation - Numerical techniques in linear algebra ◦ Matrix inverse ◦ Eigenvalue problem ◦ Diagonalisation of matrices ◦ Singular value decomposition							
UNIT-III (10 Hrs)	Numerical techniques in Probability and Statistics, (Pseudo) Random number generation , Computing statistical moments for data samples, Least Squares fitting, Error Analysis ,Hypothesis Testing, Monte Carlo sampling Applications to Quantum Mechanics (can be done using openly available modules in							

	languages like Python, Julia etc.) ○ Eigen energies of coupled two level systems Eigen energies of two-level system coupled to oscillator (Jaynes-Cummings Model) ○ Driven two-level system – Rabi Problem ○ Driven damped oscillator — coherent states
UNIT-IV (10Hrs)	Applications to EM theory (e.g. magnetic field simulation) ○ Electrostatic charge distributions ○ Magnetostatic current distributions ○ Finite Element techniques for electromagnetic simulations
Suggested Lab Experiments	
Exp. No.	Experiment Title
1.	Develop programs using classes, objects, and data structures for scientific data storage, retrieval, and memory management
2.	Implement and benchmark sorting algorithms (Bubble Sort, Merge Sort, Quick Sort) and analyze their computational efficiency.
3.	Implement searching algorithms (Linear Search and Binary Search) and compare their execution times for different data sizes.
4.	Develop programs for arithmetic computations such as Greatest Common Divisor (GCD), Prime Number Generation, and Prime Factorization.
5.	Solve second-order ordinary differential equations with constant and variable coefficients using numerical methods.
6.	Implement numerical solutions for boundary value problems such as Poisson and Laplace equations.
7.	Simulate Wave and Diffusion equations using finite difference techniques and visualize the results.
8.	Develop programs for matrix operations including matrix inversion, eigenvalue computation, and matrix diagonalization.
9.	Implement Singular Value Decomposition (SVD) and demonstrate its application in data reduction or image compression.
10.	Generate pseudo-random numbers and compute statistical moments, perform least-squares fitting, and conduct error analysis on experimental datasets.
11.	Perform Monte Carlo simulations and hypothesis testing for selected engineering or physical science applications.
12.	Develop computational models for Quantum Mechanical systems (e.g., coupled two-level systems or Rabi problem) and Electromagnetic field simulations (electrostatic or magnetostatic distributions).
REFERENCE BOOKS:	
1.	Computational Physics, Nicholas Giordano, Hisao Nakanishi, 2nd edition, Pearson Addison Wesley (2005)

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23QTM104	Minor	3	0	0	3	30	70	3 Hrs.
INTRODUCTION TO QUANTUM COMPUTATION								
(Minor Degree Course in Quantum Technologies)								
Pre-requisites: Basic Physics, Linear Algebra, and Introduction to Modern Physics.								
Course Objectives: Students are expected								
1.	Introduce the basic principles of quantum mechanics as they relate to information theory and computation.							
2.	Provide foundational knowledge of qubits, quantum gates, and quantum circuits.							
3.	Explore key quantum algorithms and contrast them with classical counterparts.							
4.	Analyze the computational complexity of quantum algorithms and understand emerging quantum computing models.							
5.	Familiarize students with quantum error correction, NISQ devices, and modern applications in quantum technologies.							
Course Out Comes: At the end of the course students will be able to								
S. No	OUT COME							Knowledge Level
1.	Describe how quantum computation works and how qubits are implemented physically .							K3
2.	Apply knowledge of quantum logic gates and circuits to basic problem-solving (K3)							K3
3.	Implement basic quantum algorithms and compare to classical methods.							K3
4.	Evaluate advanced quantum algorithms and their complexity.							K4
5.	Analyze quantum error correction techniques and emerging quantum computing technologies.							K4
SYLLABUS								
UNIT-I (10 Hrs)	Foundations of Quantum Computation: Differences between classical bits and qubits, Dirac notation, quantum states, and superposition, Bloch sphere representation, Entanglement and Bell states, No-cloning theorem, Physical implementations: spin-½ systems, photon polarization, ion traps, superconducting circuits, quantum dots (TB 1 - Ch 1 and 2; TB-2 – Ch 1 and 2)							
UNIT-II (10 Hrs)	Quantum Gates, Circuits, and Reversible Computation : Single-qubit gates: Pauli-X, Y, Z, Hadamard, Phase; Two-qubit gates: CNOT, SWAP; Circuit model of quantum computation; Reversible logic and need for reversibility in quantum computing; Solovay– Kitaev theorem (conceptual); Review of classical Turing machines; Classical							

	complexity classes: P, NP, PSPACE (Text Books - Nielsen & Chuang – Ch. 4–5 ; Manenti & Motta – Ch. 2–3)
UNIT-III (10 Hrs)	Foundational Quantum Algorithms : Quantum parallelism and its advantage, Deutsch Algorithm, Deutsch–Jozsa Algorithm, Bernstein–Vazirani Algorithm, Simon’s Algorithm, Grover’s Algorithm for unstructured search, Oracle-based problem solving (Text Book 1- Ch. 6; Text Book 2 – Ch. 4)
UNIT-IV (10 Hrs)	Advanced Algorithms and Quantum Complexity: Quantum Fourier Transform (QFT): theory and implementation, Shor’s Algorithm for prime factorization, Overview of quantum complexity classes: Q, EQP, BQP, BPP, QMA; Modern quantum algorithms: Variational Quantum Eigensolver (VQE), HHL Algorithm for linear systems, Quantum Approximate Optimization Algorithm (QAOA) (Text Book 1 – Ch. 5, 7; Text Book 2: Ch. 5, 6)
UNIT-V (10 Hrs)	Quantum Error Correction and Emerging Technologies: Quantum noise and decoherence, Bit-flip, phase-flip, and Shor code; Fault-tolerant quantum computing (conceptual overview); NISQ (Noisy Intermediate-Scale Quantum) processors; Recent developments in quantum hardware; Quantum advantage: examples and controversy; Roadmap toward large-scale quantum computation (Text Books 1 - Ch. 10, Text Book 2 – Ch. 7–8)
TEXTBOOK:	
1.	Quantum Computation and Quantum Information by Michael A. Nielsen and Isaac L. Chuang.
2.	Quantum Information Science, Riccardo Manenti & Mario Motta, Oxford University Press
REFERENCE BOOKS:	
1.	IBM Quantum Learning & Qiskit Textbook • Free online course & interactive textbook on quantum circuits, algorithms, and Python-based Qiskit coding. Qiskit Textbook : https://qiskit.org/textbook/ IBM Quantum Learning : IBM Quantum Learning
2.	Microsoft Quantum Katas • Self-paced exercises using Q#, great for learning gates, circuits, and quantum algorithms.  Quantum Katas GitHub
3.	Google Quantum AI • Tutorials, videos, and code examples using Cirq (Google’s quantum SDK).  Google Quantum AI : https://quantumai.google/resources
4.	Coursera: Quantum Computing Courses • Offers structured beginner to intermediate courses, including quantum algorithms and error correction.  Coursera – Quantum Computing
5.	YouTube: Qiskit Channel • Lectures, demos, and tutorials on real quantum circuits and algorithms.  Qiskit YouTube

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23QTM105	Minor	3	0	0	3	30	70	3 Hrs.
INTRODUCTION TO QUANTUM COMMUNICATION								
(Minor Degree Course in Quantum Technologies)								
Pre-requisites: Basic Physics, Linear Algebra, and Introduction to Modern Physics.								
Course Objectives: Students are expected								
1.	Introduce foundational optics and detection principles relevant to quantum communication.							
2.	Explore key ideas in classical and quantum information theory							
3.	Understand fundamental quantum communication protocols including teleportation and key distribution.							
4.	Analyze the role of entanglement, quantum repeaters, and quantum memories in long distance communication.							
5.	Review hardware implementations and architectures for quantum networks and the quantum internet.							
Course Out Comes: At the end of the course students will be able to								
S. No	OUT COME							Knowledge Level
1.	Demonstrate understanding of polarization optics and photo detection techniques.							K3
2.	Apply concepts of information theory in both classical and quantum settings.							K3
3.	Explain core quantum communication protocols like teleportation and QKD.							K3
4.	Analyze system-level concepts like quantum repeaters, networks, and hardware.							K4
5.	Analyze quantum network architectures and physical implementations for secure communication.							K4
SYLLABUS								
UNIT-I (10Hrs)	Fundamentals of Polarization and Detection: Polarization optics: quarter-wave and half-wave plates, Polarizing beam splitters and their role in quantum optics, Linear detectors: photodiodes, homodyne & heterodyne detection, Square-law detectors: intensity measurements, Photomultipliers and avalanche photodiodes (APDs) , (TB-1, Ch. 2, 3)							
UNIT-II (10 Hrs)	Basics of Classical Information Theory: Shannon entropy and mutual information, Noiseless source coding (Shannon’s theorem), Noisy channel capacity (basics), Relevance of classical theory to quantum communication (TB-2-Ch. 11, TB-1- Ch. 5)							
UNIT-	Principles of Quantum Communication : No-cloning theorem and its implications,							

III (10 Hrs)	Quantum teleportation protocol, Quantum dense coding, Bell's theorem and entanglement, Bell-state measurement and experimental tests, Quantum memories and quantum repeaters (TB-2, Ch. 1–2, 12; TB-1, Ch. 6)
UNIT-IV (10Hrs)	Quantum Key Distribution (QKD) Protocols: Principles of secure quantum communication, BB84, E91, BBM92, B92 protocols, Coherent One-Way (COW) and Differential Phase Shift (DPS) protocols, Role of entanglement in QKD, Error correction and privacy amplification (introductory), (TB-1, Ch. 7; TB-2, Ch. 12)
UNIT-V (10Hrs)	Quantum Networks and Physical Implementations: Quantum internet concepts and architecture, Role of quantum repeaters in extending communication range, Survey of Hardware implementations: Free-space optics, Satellite-based QKD (e.g., Micius satellite), Fiber-optic based QKD networks; Challenges in large-scale deployment and synchronization (TB-1, Ch. 8)
TEXTBOOKS:	
1.	Elements of Quantum Computation and Quantum Communication, A. Pathak, CRC Press
2.	Quantum Computation and Quantum Information, M. Nielsen & I. Chuang, Cambridge Univ. Pre
Suggested E-Content & Resources:	
1.	Qiskit Textbook – Quantum Communication Section
2.	Quantum Internet Alliance
3.	Microsoft Quantum Learning
4.	Quantum World YouTube Lectures – Quantum Key Distribution